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Asymmetric Catalysis

Iridium-Catalyzed Enantioselective Formal *meta*-Allenylation of Pyridines

Pankaj Roy[+], *Pratibha Mahto*[+], *and Santanu Mukherjee**

Abstract: Site-selective functionalization of pyridines is an important yet challenging transformation. Temporary dearomatization of pyridines through redox-neutral cycloaddition has recently emerged as a reliable and attractive strategy for *meta*-functionalization of pyridines. Despite the amenability of the resulting bench-stable cycloadduct, oxazinopyridine, to a variety of electrophilic and radical-based reagents, its enantioselective functionalization remains unexplored. Herein, we present the first enantioselective *meta*-functionalization of pyridines through the intermediacy of oxazinopyridines. This highly enantioselective formal *meta*-allenylation of pyridines is achieved by treating the easily accessible oxazinopyridines with racemic allenylic alcohol under the combination of a catalytic amount of an Ir(I)/(P, olefin) complex and Lewis acidic $Sc(OTf)_3$ followed by acid-mediated rearomatization. With oxazinopyridines derived from unsubstituted or suitably substituted pyridines, iterative 3,5-difunctionalization is possible, which also allows for the stereodivergent synthesis of all four stereoisomers of 3,5-bis-allenylic pyridines.

Introduction

Pyridine is among the most abundant nitrogenous heterocycles in natural products,[1] pharmaceuticals,[2–4] and agrochemicals[5] (Figure 1). Pyridine derivatives also serve as ligands[6] and catalysts[7,8] in asymmetric synthesis besides acting as precursors to targets of biological and material importance.[9] Consequently, site-selective functionalization of pyridines is a highly sought-after transformation. Direct C–H functionalization is the most straightforward avenue of decorating pyridines.[10,11] The inherent electron-deficient nature of pyridines renders their *ortho*- and *para*-positions electrophilic and amenable to nucleophilic substitution (Scheme 1a).[12,13] In contrast, *meta*-functionalization of pyridines is challenging.[14–16] Electrophilic aromatic substitution reactions such as nitration[17] and bromination[18] of pyridines are known but require harsh reaction conditions, which restrict their application.

The adversities associated with functionalizing the *meta*-position of pyridines have spurred innovative strategies in recent years.[14,15] Various approaches, including directing group-assisted C–H activation,[19,20] non-directed metalation,[21–23] and temporary dearomatization of the pyridine ring,[24,25] have been explored to enable *meta*-functionalization, of which the latter emerged as the most attractive one (Scheme 1b). Among different temporary dearomatization strategies adopted so far,[24,26–31] the redox-neutral dearomatization of pyridine through cycloaddition is particularly noteworthy.

In 1967, Huisgen et al. reported a three-component cycloaddition of pyridine or isoquinoline with dimethyl acetylenedicarboxylate (DMAD) and phenyl isocyanate, which could be converted back to its components upon heating.[32] The Nair group further advanced this area by substituting phenyl isocyanate with other dipolarophiles, thereby expanding the scope of this reaction.[33–36]

Notwithstanding the reversibility of this cycloaddition reaction, its potential in functionalizing pyridines was not realized until recently. In 2022, Studer et al. exploited this strategy to transform pyridines into bench-stable dearomatized oxazinopyridines **C**, via Huisgen 1,4-dipolar cycloaddition with DMAD and methyl pyruvate (Scheme 1b).[28] These oxazinopyridines (oxazinoazines) are nucleophilic in nature by virtue of their (di)enamine character and may be viewed as *meta*-nucleophilic pyridine surrogates. Consequently, they have been intercepted with a variety of electrophiles and radical-based reagents prior to rearomatization to afford *meta*-substituted pyridines.[28,37–41] In addition, *meta*-sulfonylation of pyridines has also been achieved under electrochemical[42] or photochemical conditions.[43,44]

Despite their growing popularity, the application of these *meta*-nucleophilic pyridine surrogates in catalytic transformations is rare.[40,43] Moreover, this redox-neutral dearomatization strategy has never been applied for enantioselective functionalization of pyridines or any other azines. In fact, the only examples of enantioselective *meta*-functionalization of pyridines, reported by Wang et al., involve reductive dearomatization of pyridines.[45,46] These processes comprise of borane-catalyzed dearomative hydroboration of pyridines to 1,4-dihydropyridines (**A**, Scheme 1b) followed by Pd- or Ir-catalyzed enantioselective *meta*-allylation and oxidative rearomatization.

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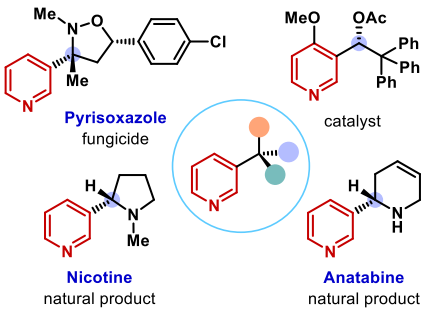


Figure 1. Examples of *meta*-functionalized pyridines bearing an adjacent stereocenter.

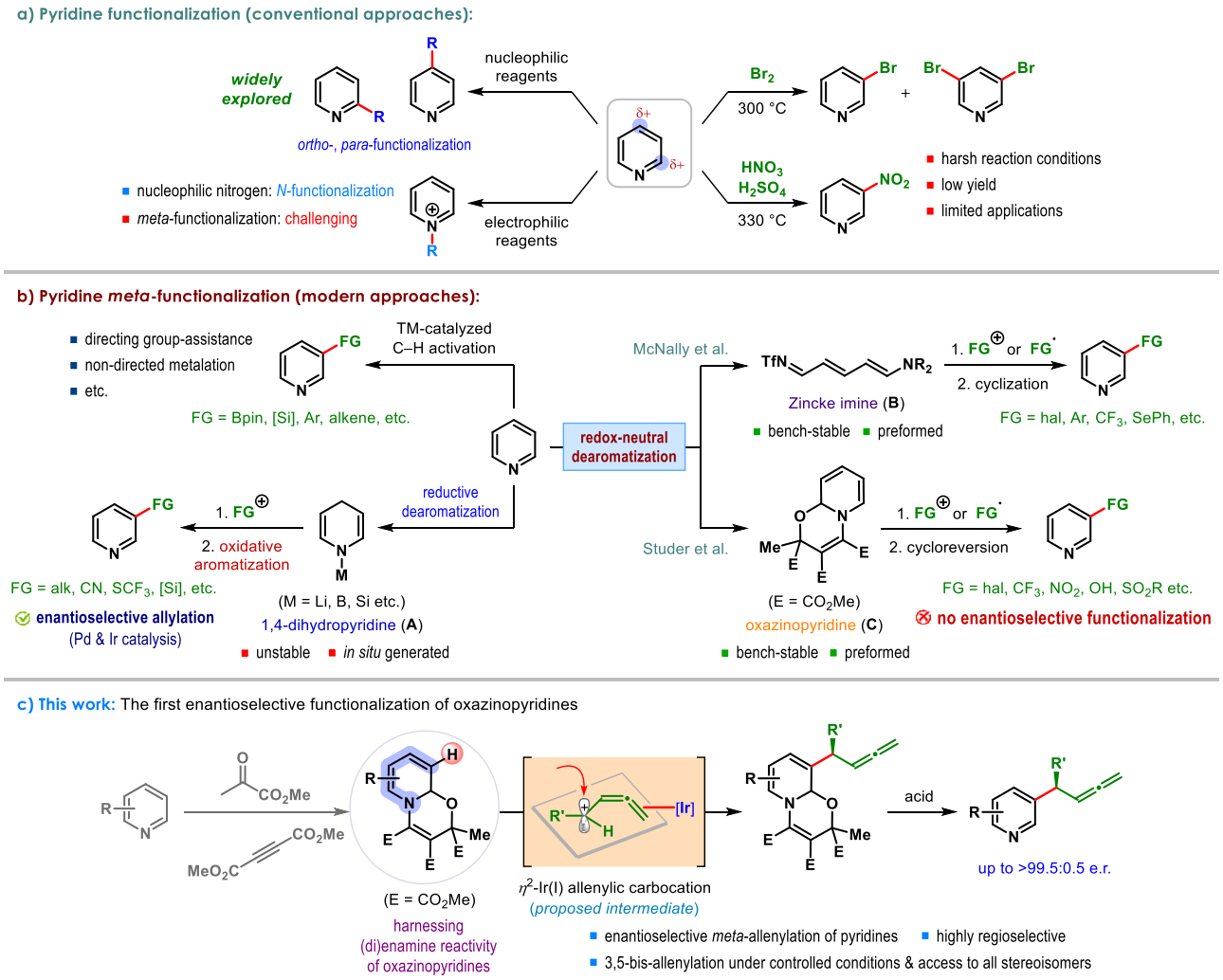
In an effort to expand the synthetic toolbox for enantioselective functionalization of pyridines with precise *meta*-regioselectivity, we herein report the first enantioselective functionalization of oxazinopyridines *en route* to chiral *meta*-functionalized pyridines.[47]

As a part of our research program on enantioselective C(sp^2)–H allenylation under iridium catalysis,[48,49] we became interested in introducing an allenylic substituent at the *meta*-position of pyridines. Ir-catalyzed allenylic substitution reaction was first reported by Takeuchi et al. in 2004.[50] However, Ir-catalyzed enantioselective allenylic substitution was pioneered by the Carreira group and was proposed to proceed via an Ir-bound allenylic carbocation intermediate.[51–53] Recent studies by our laboratory[48] and the Takeuchi group[54] demonstrated the utility of this catalyst system for Friedel–Crafts allenylation of (hetero)arenes. However, electron-deficient nature of pyridines precludes their direct application in Friedel–Crafts allenylation.

We hypothesized that the nucleophilic (di)enamine functionality of oxazinopyridines[28] can be intercepted with an Ir-bound electrophilic allenylic fragment[53] (Scheme 1c). Subsequent acid-mediated rearomatization would furnish chiral pyridine derivatives, bearing a synthetically versatile allenylic group at the *meta*-position.

Results and Discussion

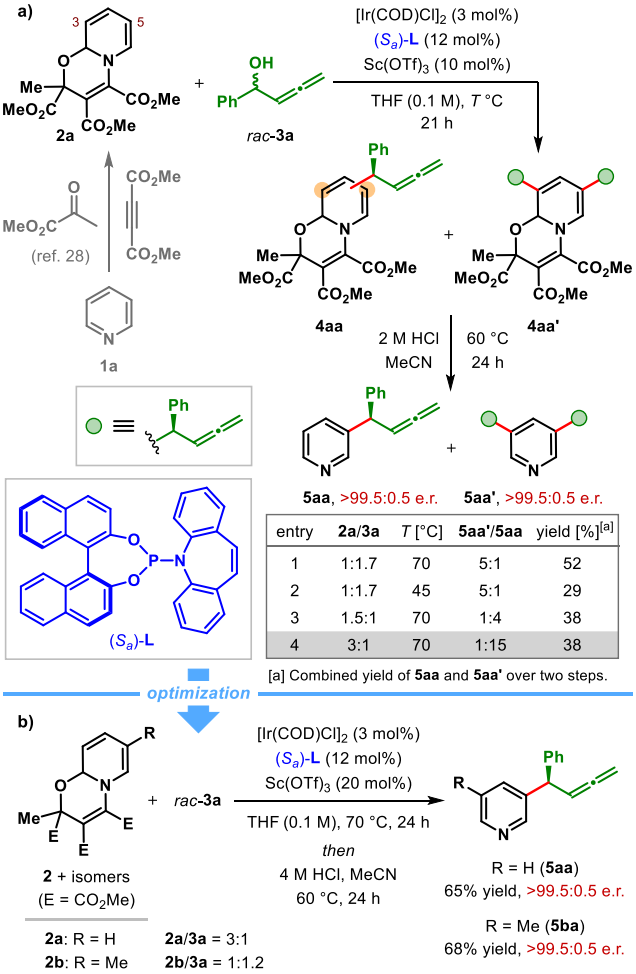
We embarked upon testing our hypothesis by selecting oxazinopyridine **2a**, derived through dearomatization of



Scheme 1. Enantioselective *meta*-allenylation of pyridines: The backdrop of this study.

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entry	2a/3a	T [°C]	5aa'/5aa	yield [%][a]
1	1:1.7	70	5:1	52
2	1:1.7	45	5:1	29
3	1.5:1	70	1:4	38
4	3:1	70	1:15	38

[a] Combined yield of **5aa** and **5aa'** over two steps.

Scheme 2. a) Initial studies and b) optimized protocols for *meta*-allenylation of pyridines.

pyridine (**1a**), as the model substrate (Scheme 2a).[28] Both C3 and C5 positions of **2a** are nucleophilic owing to their dienamine and enamine reactivity, respectively, which gives rise to the possibility of over-allenylation. Therefore, it was imperative to identify conditions for selective mono-allenylation of **2a** at the outset of our investigation. The Ir(I)-complex derived in situ from a 1:2 mixture of $[Ir(COD)Cl]_2$ and Carreira's (*P*, olefin) ligand[55] (S_a)-**L** was selected as the catalyst candidate. A reaction of **2a** with an excess (1.7 equiv) of racemic allenylic alcohol **3a**, in the presence of 6 mol% of the Ir(I)/(S_a)-**L** complex and 10 mol% of $Sc(OTf)_3$ at 70 °C, indeed furnished bis-allenylic oxazinopyridine **4aa′** predominantly in an overall 58% yield within 21 h. The exact position of the allenylic substituent in **4aa** (i.e., C3 or C5), although inconsequential for the final outcome, could not be determined. Treatment of this mixture of **4aa** and **4aa′** with 2 M HCl in acetonitrile at 60 °C led to rearomatization and afforded a 1:5 mixture of 3-allenylic pyridine (**5aa**) and 3, 5-bis-allenylic pyridine (**5aa′**) in 90% yield essentially as a single enantiomer for each of them (Scheme 2a, entry 1). Lowering the temperature to 45 °C in an attempt to prevent over-allenylation proved futile, as **5aa′** still formed as the major product (Scheme 2a, entry 2). However, reversing the stoichiometry of **2a** and **3a** reduced the extent of bis-allenylation considerably. Finally, a 3:1 ratio of **2a** and **3a** turned out to be the optimum and favored mono-allenylation (**5aa/5aa′** 15:1). Under these conditions, **5aa** was isolated in an overall 36% yield with >99:1 e.r. (Scheme 2a, entry 4).

Starting with these initial conditions for selective mono-allenylation of **2a** *en route* to *meta*-allenylic pyridine **5aa**, we undertook a detailed optimization of various reaction parameters for improving the yield of **5aa** (see Supporting Information). These variables included promoter, reaction medium, temperature, and initial concentration for the allenylation step as well as the choice of acid and temperature for the rearomatization step.

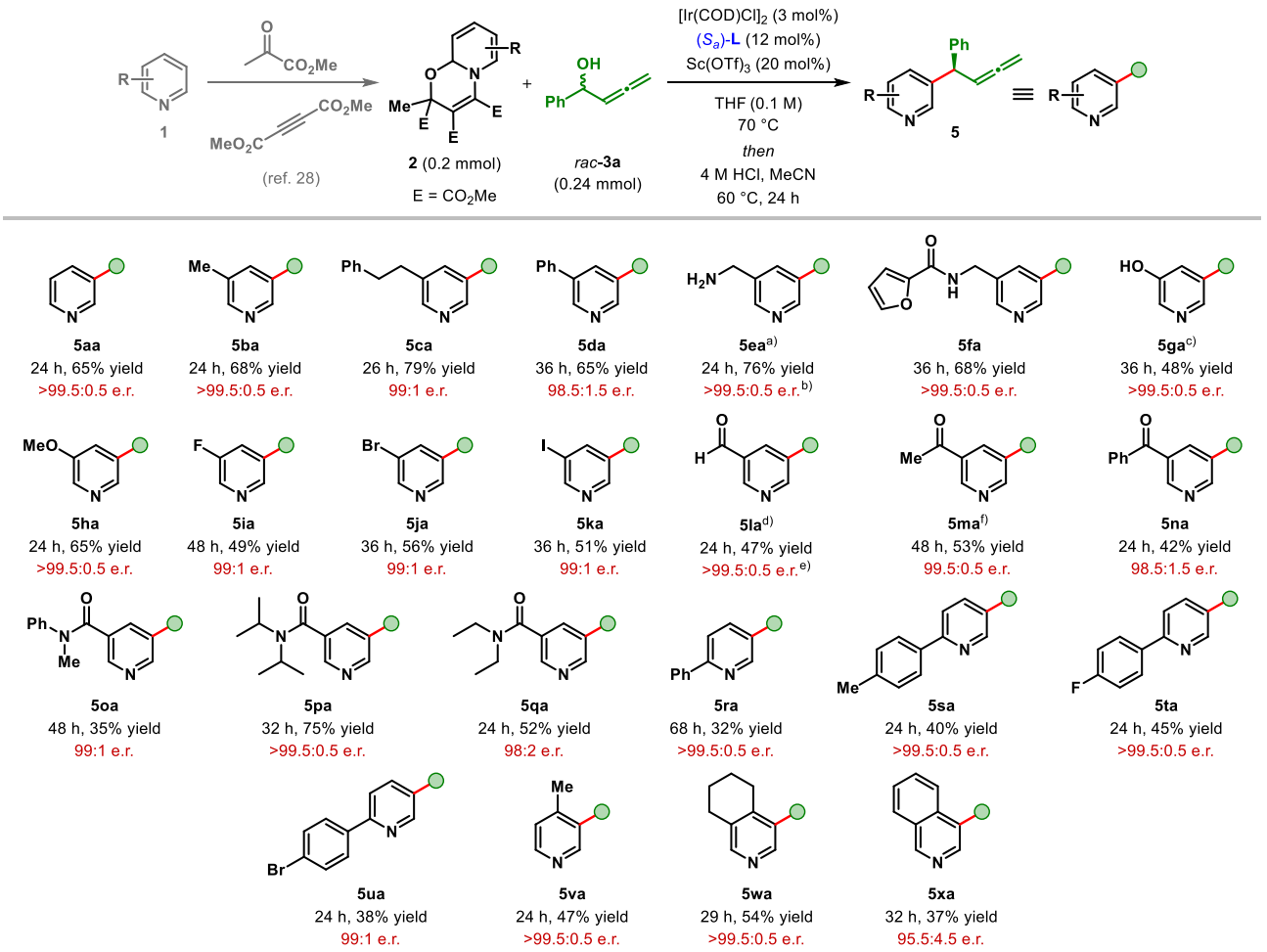
These studies revealed a combination of 6 mol% of the Ir(I)/(S_a)-**L** complex and 20 mol% of $Sc(OTf)_3$ to be the optimum for the allenylation reaction at 70 °C in THF (Scheme 2b). Subsequent rearomatization with 4 M HCl in acetonitrile at 60 °C afforded *meta*-allenylic pyridine **5aa** with >99.5:0.5 e.r. in 65% yield over two steps. It is worth noting that oxazinopyridine **2a** contains two stereocenters and forms as an inseparable mixture of diastereomers (with 4.5:1 d.r.). However, the stereochemistry of **2a** did not seem to have any effect on the final stereochemical outcome, since **5aa** was isolated with outstanding e.r. The enantioinduction of the allenylation reaction was completely controlled by the Ir(I)/(S_a)-**L** complex.

Similarly, 3-substituted pyridines can give rise to two regioisomeric oxazinopyridines, which are inseparable. However, each of these oxazinopyridines contains only one accessible position for allenylation and upon rearomatization should converge into the same product. As expected from the reaction with **2a**, catalyst controlled-enantioinduction should override the substrate control. Indeed, when oxazinopyridine **2b**, derived from 3-methyl pyridine **1b**, was reacted as a mixture of diastereo- and regioisomers in a 1:1.2 ratio with *rac*-**3a**, the 3-allenylic 5-methyl pyridine **5ba** was isolated in 68% yield with >99.5:0.5 e.r. (Scheme 2b). Clearly, each isomer of **2b** yields the same product enantiomer upon rearomatization with equally high enantioselectivity.

Having optimized the reaction conditions, the scope and limitations of *meta*-allenylation of pyridines were investigated. We first sought to examine the scope of pyridines in this reaction (Scheme 3). A variety of oxazinopyridines (**2b-q**), derived from 3-substituted pyridines, reacted efficiently with allenylic alcohol *rac*-**3a** under our optimized reaction conditions. In all these cases, the isomeric mixtures of oxazinopyridines were employed as substrate, and the corresponding *meta*-allenylic pyridines **5ba-qa** were obtained in moderate to good yields over two steps with uniformly outstanding e.r. Electron-rich oxazinopyridines (**2b-h**) turned out to be superior substrates compared to most of the electron-deficient ones (**2i-q**), as the products with relatively higher yields were generally observed with the former. Protected functionalities on oxazinopyridines such as NHBoc (**2e**), OAc (**2g**), acetal (**2l**) didn't hamper the allenylation reaction, but the protecting groups fell off under the acidic conditions employed for the rearomatization step. However, we were pleased to note that other functional groups such as amide and ketone survived all three steps.

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Scheme 3. Scope of azines in enantioselective formal *meta*-allenylation. Unless noted otherwise, the reaction conditions indicated above were followed. Yields refer to the isolated product, over two steps starting from an isomeric mixture of oxazinopyridines after chromatographic purification. Enantiomeric ratios (e.r.) were determined by HPLC analysis using a stationary-phase chiral column. [a] Starting from the corresponding *tert*-butyl carbamate. [b] E.r. of the corresponding *tert*-butyl carbamate. [c] Starting from the corresponding acetate. [d] Starting from the corresponding ethylene glycol acetal. Reaction with **2l** (0.6 mmol) and **3a** (0.2 mmol). [e] E.r. of the corresponding alcohol. [f] Reaction with **2m** (0.6 mmol) and **3a** (0.2 mmol).

In the case of 2-substituted pyridines, availability of both 3- and 5-positions gives rise to the possibility of regioisomeric allenylation apart from bis-allenylation. Interestingly, oxazinopyridine **2r**, derived from 2-phenyl pyridine, did not pose any regioselectivity issue and furnished 2-phenyl 5-allenylic pyridine **5ra** exclusively with excellent enantioselectivity, albeit in only 32% yield. The same trend was observed with other 2-aryl pyridine-derived oxazinopyridines (**2s-u**). In these instances, allenylation proceeded through the dienamine functionality of the oxazinopyridine to avoid steric repulsion with the aryl substituent. To the best of our knowledge, this is the first example of enantioselective C5-functionalization of 2-aryl pyridines. Unfortunately, 2-alkyl pyridines failed to react in the Huisgen 1,4-dipolar cycloaddition step to generate oxazinopyridines and, therefore, aren't compatible with this dearomative *meta*-allenylation strategy. 2-Heteroaryl pyridines furnished the desired product but in low yield (see Supporting Information).

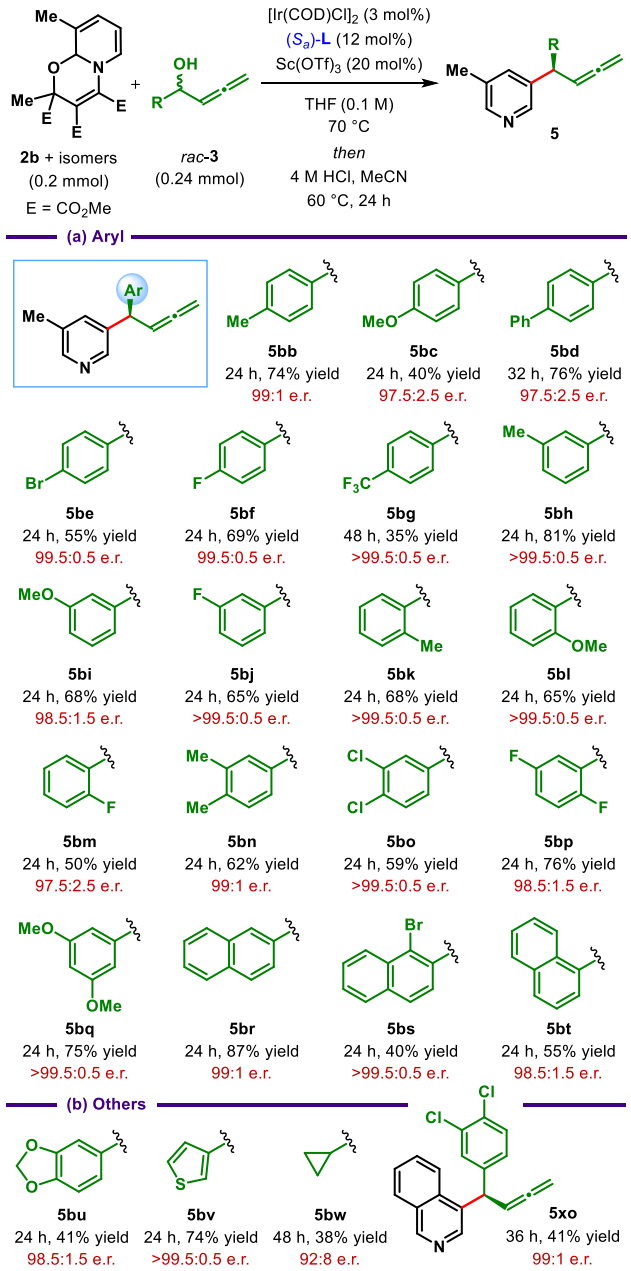
4-Methyl pyridine-derived oxazinopyridine **2v** is also amenable to allenylation under our optimized reaction conditions and led only to mono-allenylation: The 3-allenylic 4-methyl pyridine **5va** was isolated in moderate yield with high enantiopurity. 5,6,7,8-Tetrahydroisoquinoline (**1w**) via dearomative *meta*-allenylation gave access to polysubstituted pyridine **5wa** in decent yield.

Subsequently, we became interested in exploring other structurally related azines, such as quinoline, isoquinoline, etc., under our protocol. While the dearomatized intermediate derived from quinoline did not participate in the allenylation reaction, isoquinoline (**1x**) smoothly produced 4-allenylic isoquinoline **5xa** in 37% yield with 95.5:4.5 e.r.

After showcasing the scope of azines, we turned to explore the scope of allenylic alcohols (Scheme 4). For this purpose, oxazinopyridine **2b**, derived from 3-methyl pyridine (**1b**), was selected as the nucleophile. An array of electron-rich (**3b-d**) and electron-deficient (**3e-g**) *para*-substituted aryl allenylic alcohols underwent successful substitution reaction with 3-methyl oxazinopyridine (**2b**) and delivered the 3-allenylic 5-methyl pyridines (**5bb-bg**) in moderate to good yields with uniformly high enantioselectivity. In addition,

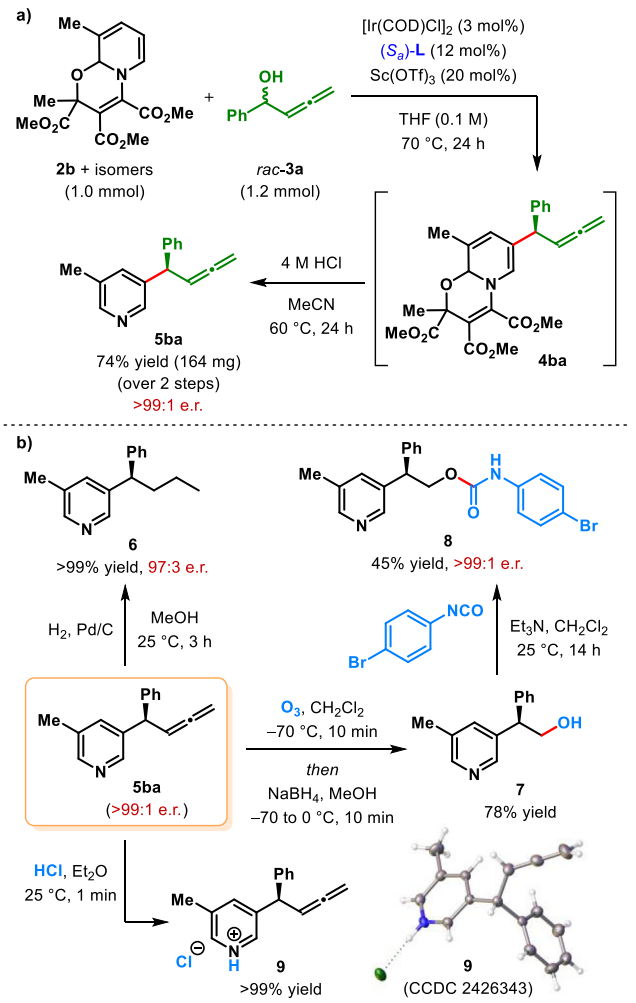
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Scheme 4. Scope of allenylic alcohols. Unless noted otherwise, the reaction conditions indicated above were followed. Yields refer to the isolated product over two steps, starting from an isomeric mixture of oxazinopyridines, after chromatographic purification. Enantiomeric ratios (e.r.) were determined by HPLC analysis using a stationary phase chiral column.

meta- and *ortho*-substituted allenylic alcohols (**3h-m**) were found to be equally competent electrophiles for this reaction and afforded the corresponding pyridine derivatives **5bh-bm** in good to high yields and excellent e.r. Electronic as well as steric diversity was equally accommodated across a spectrum of disubstituted aryl rings on the allenylic alcohols (**3n-q**). Polyaromatics (**3r-t**) and heterocyclic group (**3u-v**) on allenylic alcohols also proved to be suitable substrates for this reaction. To our delight, cyclopropyl-substituted allenylic



Scheme 5. a) Scale-up synthesis and b) synthetic modifications of **5ba**.

alcohol **3w** furnished the 3-allenylic pyridine **5bw** in 38% yield with 92:8 e.r.

Unfortunately, products derived from alkenyl and alkynyl substituted allenylic alcohols were isolated in poor yields under our reaction conditions (see Supporting Information), which remains a limitation of the current protocol.

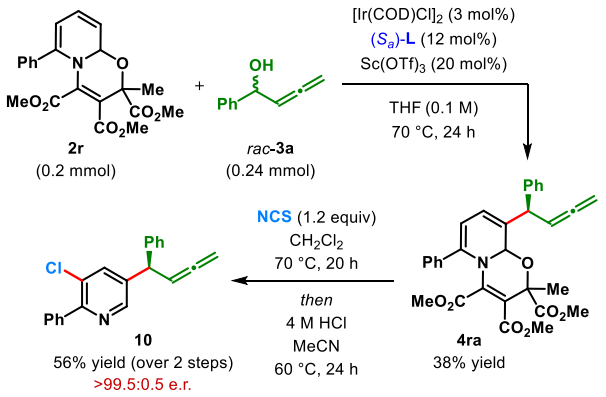
As expected, other combinations of oxazinoazine derivatives with allenylic alcohols can be employed in this Ir-catalyzed reaction. This aspect was demonstrated by allenylation of isoquinoline with 3,4-dichlorophenyl-substituted allenylic alcohol **3o**. The resulting 4-allenylic isoquinoline **5xo** was isolated with 99:1 e.r. in 41% yield over two steps (Scheme 4b).

To showcase the practicality of this protocol, a 1.0 mmol scale reaction was carried out between branched allenylic alcohol *rac*-**3a** and 3-methyl oxazinopyridine **2b**, which afforded **5ba** in 74% yield with a similar level of enantioselectivity (Scheme 5a).

The utility of the allene functionality was demonstrated through a few synthetic elaborations involving **5ba** (Scheme 5b). Hydrogenation of the allene functionality in **5ba** under catalytic Pd/C led to the formation of 3,5-dialkylated pyridine **6** in quantitative yield. Reductive

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Scheme 6. Sequential difunctionalization of 2-phenyl pyridine.

ozonolysis of **5ba** produced primary alcohol **7** in 78% yield, which was converted to carbamate **8** by reacting with *p*-bromophenylisocyanate.

Even though we failed to crystallize the allenylic pyridines shown in Schemes 3 and 4 or the derivatives (**6–8**) shown in Scheme 5, the hydrochloride salt of **5ba**, prepared by treating it with a solution of HCl in Et_2O, formed diffraction quality crystals. Single crystal X-ray diffraction analysis of the pyridinium chloride **9** revealed the absolute configuration of **5ba** as (*R*) (CCDC 2426343, Scheme 5b).[56] The absolute configuration of other pyridine derivatives reported in this article was assigned in analogy with **5ba**.

Given the prevalence of multifunctional heterocycles in bioactive molecules,[2–4] systematic installation of functionalities around the pyridine core in a site-selective manner is an alluring goal. However, such a transformation, especially involving an enantioselective step, is elusive. Exclusive C3-selective allenylation of 6-phenyl oxazinopyridine (**2r**) intrigued us to assess the possibility of introducing a less sterically demanding electrophile adjacent to the phenyl group following enantioselective allenylation. Accordingly, when 3-allenylic 6-phenyl oxazinopyridine **4ra** was subjected to a reaction with *N*-chlorosuccinimide (NCS) in CH_2Cl_2 at 70 °C,[28] chlorination through the enamine functionality of **4ra** indeed took place to yield 5-allenylic-3-chloro-2-phenylpyridine **10** after rearomatization (Scheme 6).

During the optimization studies (Scheme 2a), bis-allenylation was encountered with unsubstituted oxazinopyridine **2a**. Using 2.2 equiv of *rac*-**3a** in this reaction, C_2-symmetric 3,5-bis-allenylic pyridine **5aa′** was isolated with excellent relative (>20:1 d.r.) and absolute (>99.5:0.5 e.r.) stereocontrol (Scheme 7a). The absence of *meso*-product indicates complete ligand control in the second allenylation and opens avenues for stereodivergent[57] iterative bis-allenylation of pyridine with two different allenylic alcohols.

To achieve this goal, controlled mono-allenylation was performed using an excess of oxazinopyridine **2a** for the reaction with 4-fluorophenyl allenylic alcohol *rac*-**3f** (Scheme 7b). Under our otherwise standard reaction conditions, both stereoisomers of **4af** were isolated in good yield. The second allenylation of **4af** with 2-naphthyl allenylic alcohol *rac*-**3r** and subsequent rearomatization afforded all four possible stereoisomers of differently substituted 3,5-bis-allenylic pyridine **11** with superb diastereo- and enantioselectivity. Overall, starting from **2a**, stereodivergent synthesis of **11** was possible simply through the choice of the ligand antipode in each step.

Based on the literature reports[52,53] and our prior experimental findings,[48] a tentative catalytic cycle is depicted in Scheme 8. Activation of allenylic alcohol (*rac*-**3**) through the η^2-coordination of its terminal double bond to Ir(I)/(*P*, olefin) complex (**3** → **12**) followed by Lewis acid-mediated ionization of the C─O bond is believed to generate Ir(I)-bound allenylic carbocation **13**.[53] Due to the steric crowding at the *si*-face of the η^2-Ir(I)-bound planar butadienylium fragment in **13** (R = Ph), caused by the binaphthyl moiety of the axially bound (S_a)-**L** on Ir(I), nucleophilic attack of oxazinopyridine **2a** is expected to occur at its sterically more accessible *re*-face. In this enantiodetermining nucleophilic addition, the C─C bond formation can take place either from the C3 or C5 position of **2a**, which would give rise to a mixture of allenylic oxazinopyridines (only 3-allenylic oxazinopyridine **4** is shown here) after deprotonation and decomplexation. For unsubstituted oxazinopyridine **2a** and oxazinopyridine derived from 4-substituted pyridines, this mixture is inconsequential since acid-mediated rearomatization would generate 3-allenylic pyridine (**5**) and 3-allenylic 4-substituted pyridine, respectively. In the case of 4-methyl oxazinopyridine (**2v**), ^{1}H NMR analysis of the unpurified intermediate revealed the incorporation of the allenylic fragment at its dienamine position (i.e., at C3) (see Supporting Information). Since the formation or the absence of the other isomer (i.e., at C5) could not be confirmed from this NMR analysis, the possibility of allenylation through the enamine position of oxazinopyridines cannot be ruled out at this stage. For 3-substituted pyridines, two regioisomeric oxazinopyridines are possible. However, each of them contains only one accessible position for allenylation and upon rearomatization converges into the same product. In contrast, non-selective nucleophilic attack of oxazinopyridine derived from 2-substituted pyridine could generate a regioisomeric mixture of allenylic pyridine derivatives. However, as shown with 6-aryl oxazinopyridines **2r-u**, only 2-aryl 5-allenylic pyridines (**5ra-ua**) were produced exclusively, arising out of the preferential dienamine reactivity of oxazinopyridines (Scheme 3). Therefore, enantioselective synthesis of regioisomeric 2-substituted 3-allenylic pyridine remains a challenge.

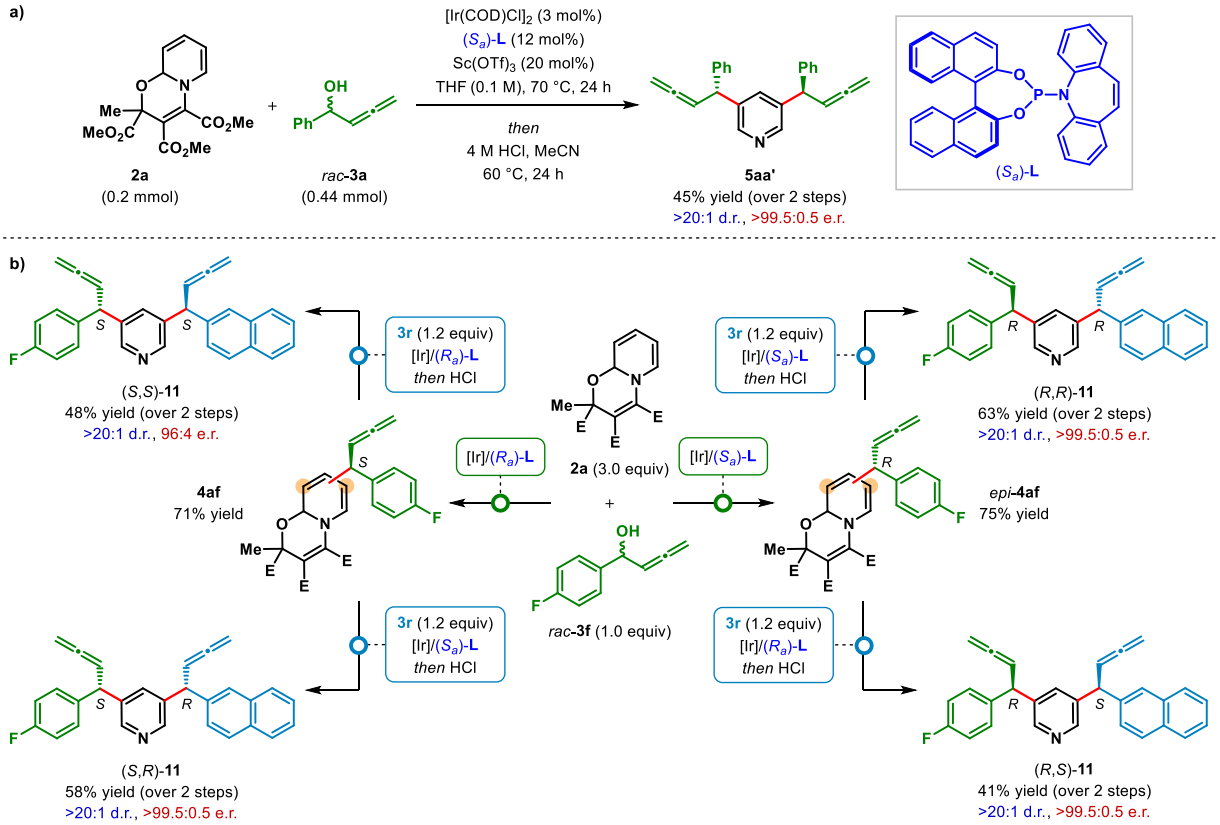
It must be noted that the oxazinopyridines contain two stereocenters and are used as a mixture of diastereomers in racemic form. While an inherent substrate bias over stereoinduction in the allenylation step gives rise to the likelihood of a match/mismatch scenario, powerful catalyst control rules out such a possibility, and *meta*-allenylic pyridines are obtained with high e.r. in each case.

Conclusion

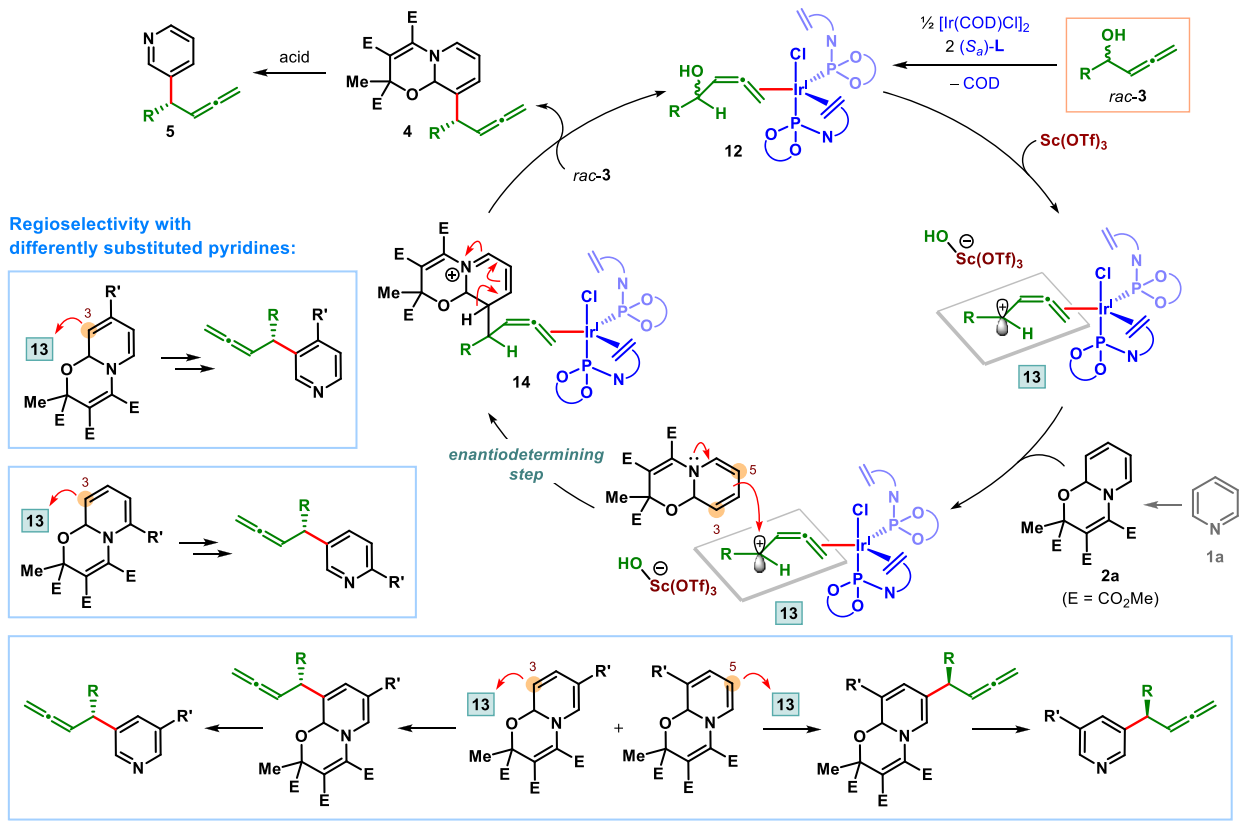
In conclusion, we have accomplished enantioselective *meta*-allenylation of pyridines via their temporary redox-neutral dearomatization to easily accessible oxazinopyridines. The nucleophilic reactivity of oxazinopyridines through their dienamine-like structural feature was leveraged to achieve

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Scheme 7. a) Bis-allenylation of pyridine and b) stereodivergent synthesis of 3,5-bis-allenylic pyridines.



Scheme 8. A tentative catalytic cycle for enantioselective formal *meta*-allenylation of pyridine.

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exclusive site-selectivity. With allenylic alcohols as electrophile, these reactions are catalyzed by a combination of Ir(I)/(*P*, olefin) complex and Lewis acidic $Sc(OTf)_3$, and furnished *meta*-allenylic pyridines, after rearomatization under acidic conditions, in moderate to good yields and generally with excellent enantioselectivity. With oxazinopyridines derived from unsubstituted or suitably substituted pyridines, iterative 3,5-difunctionalization is possible, which also allowed for the stereodivergent synthesis of 3,5-bis-allenylic pyridines with exquisite diastereo- and enantioselectivities. This is the first example of the use of oxazinopyridines in an enantioselective transformation.

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Conflict of Interests

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available in the supplementary material of this article.

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[1] D. O'Hagan, *Nat. Prod. Rep.* **2000**, *17*, 435–446.
[2] C. M. Marshall, J. G. Federice, C. N. Bell, P. B. Cox, J. T. Njardarson, *J. Med. Chem.* **2024**, *67*, 11622–11655.
[3] P. Bhutani, G. Joshi, N. Raja, N. Bachhav, P. K. Rajanna, H. Bhutani, A. T. Paul, R. Kumar, *J. Med. Chem.* **2021**, *64*, 2339–2381.
[4] E. Vitaku, D. T. Smith, J. T. Njardarson, *J. Med. Chem.* **2014**, *57*, 10257–10274.
[5] V. V. Zakharychev, A. M. Martsynkevich, *Adv. Agrochem.* **2025**, *4*, 30–48.
[6] H.-L. Kwong, H.-L. Yeung, C.-T. Yeung, W.-S. Lee, C.-S. Lee, W.-L. Wong, *Coord. Chem. Rev.* **2007**, *251*, 2188–2222.
[7] R. Kluga, A. Kinens, E. Suna, *Chem. Eur. J.* **2024**, *30*, e202301136.
[8] R. Gujjarappa, N. Vodnala, C. C. Malakar, *ChemistrySelect* **2020**, *5*, 8745–8758.
[9] G. de Ruiter, M. Lahav, M. E. van der Boom, *Acc. Chem. Res.* **2014**, *47*, 3407–3416.
[10] S. B. Mohite, Y. K. Mirza, P. S. Bera, S. Nadigar, S. Yugendhar, R. Karpoormath, M. Bera, *Chem. Eur. J.* **2025**, *31*, e202403032.
[11] S. Maity, A. Bera, A. Bhattacharjya, P. Maity, *Org. Biomol. Chem.* **2023**, *21*, 5671–5690.
[12] F.-Y. Zhou, L. Jiao, *Synlett* **2020**, *32*, 159–178.
[13] K. Murakami, S. Yamada, T. Kaneda, K. Itami, *Chem. Rev.* **2017**, *117*, 9302–9332.
[14] H. Cao, Q. Cheng, A. Studer, *Angew. Chem. Int. Ed.* **2023**, *62*, e202302941.
[15] S. Chakraborty, A. T. Biju, *Angew. Chem. Int. Ed.* **2023**, *62*, e202300049.
[16] D. E. Stephens, O. V. Larionov, *Tetrahedron* **2015**, *71*, 8683–8716.
[17] H. Suzuki, M. Iwaya, T. Mori, *Tetrahedron Lett.* **1997**, *38*, 5647–5650.
[18] S. M. E. Englert, S. M. McElvain, *J. Am. Chem. Soc.* **1929**, *51*, 863–866.
[19] M. Wasa, B. T. Worrell, J.-Q. Yu, *Angew. Chem. Int. Ed.* **2010**, *49*, 1275–1277.
[20] G. C. Clososki, C. J. Rohbogner, P. Knochel, *Angew. Chem. Int. Ed.* **2007**, *46*, 7681–7684.
[21] T. Zhang, Y.-X. Luan, N. Y. S. Lam, J.-F. Li, Y. Li, M. Ye, J.-Q. Yu, *Nat. Chem.* **2021**, *13*, 1207–1213.
[22] L. Yang, N. Uemura, Y. Nakao, *J. Am. Chem. Soc.* **2019**, *141*, 7972–7979.
[23] M. Ye, G.-L. Gao, J.-Q. Yu, *J. Am. Chem. Soc.* **2011**, *133*, 6964–6967.
[24] J. Lei, Z. G. Xu, *Mol. Divers.* **2025**, *29*, 849–869.
[25] C.-S. Giam, S. D. Abbott, *J. Am. Chem. Soc.* **1971**, *93*, 1294–1296.
[26] G.-Q. Sun, P. Yu, W. Zhang, W. Zhang, Y. Wang, L.-L. Liao, Z. Zhang, L. Li, Z. Lu, D.-G. Yu, S. Lin, *Nature* **2023**, *615*, 67–72.
[27] Z. Liu, J.-H. He, M. Zhang, Z.-J. Shi, H. Tang, X.-Y. Zhou, J.-J. Tian, X.-C. Wang, *J. Am. Chem. Soc.* **2022**, *144*, 4810–4818.
[28] H. Cao, Q. Cheng, A. Studer, *Science* **2022**, *378*, 779–785.
[29] B. T. Boyle, J. N. Levy, L. de Lescure, R. S. Paton, A. McNally, *Science* **2022**, *378*, 773–779.
[30] S. Wübbolt, M. Oestreich, *Angew. Chem. Int. Ed.* **2015**, *54*, 15876–15879.
[31] O. Tsuge, S. Kanemasa, T. Naritomi, J. Tanaka, *Chem. Lett.* **1984**, *13*, 1255–1258.
[32] R. Huisgen, M. Morikawa, K. Herbig, E. Brunn, *Chem. Ber.* **1967**, *100*, 1094–1106.
[33] V. Nair, A. Deepthi, D. Ashok, A. E. Raveendran, R. R. Paul, *Tetrahedron* **2014**, *70*, 3085–3105.
[34] V. Nair, A. R. Sreekanth, N. Abhilash, A. T. Biju, B. R. Devi, R. S. Menon, N. P. Rath, R. Srinivas, *Synthesis* **2003**, 1895–1902.
[35] V. Nair, A. R. Sreekanth, A. T. Biju, N. P. Rath, *Tetrahedron Lett.* **2003**, *44*, 729–732.
[36] V. Nair, A. R. Sreekanth, N. Abhilash, M. M. Bhadbhade, R. C. Gonnade, *Org. Lett.* **2002**, *4*, 3575–3577.
[37] K. Balanna, A. Studer, *J. Am. Chem. Soc.* **2025**, *147*, 7485–7495.
[38] D. Bhattacharya, A. Studer, *Angew. Chem. Int. Ed.* **2025**, *64*, e202423512.
[39] M. Haring, K. Balanna, Q. Cheng, J. Lammert, A. Studer, *J. Am. Chem. Soc.* **2024**, *146*, 30758–30763.
[40] S.-M. Guo, P. Xu, A. Studer, *Angew. Chem. Int. Ed.* **2024**, *63*, e202405385.
[41] P. Xu, Z. Wang, S.-M. Guo, A. Studer, *Nat. Commun.* **2024**, *15*, 4121.
[42] S. Qin, M. Yang, M. Xu, Z.-H. Peng, J. Cai, S. Wang, H. Gao, Z. Zhou, A. S. K. Hashmi, W. Yi, Z. Zeng, *Nat. Commun.* **2024**, *15*, 7428.
[43] M. Yang, M. Wang, H. Zhang, S. Qin, M. Zaman, J. Luo, S. Zhan, X. Wang, C. Zhang, S. Wang, H. Gao, Z. Zhou,

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A. S. K. Hashmi, W. Yi, Z. Zeng, *Chem Catal.* **2025**, *5*, 101326.
[44] Y. Ye, Z. Ye, M. Guo, X. He, C. Huang, H. Xiang, K. Chen, H. Yang, *Chem. Commun.* **2025**, *61*, 6368–6371.
[45] Z. Liu, Z.-J. Shi, L. Liu, M. Zhang, M.-C. Zhang, H.-Y. Guo, X.-C. Wang, *J. Am. Chem. Soc.* **2023**, *145*, 11789–11797.
[46] J.-J. Tian, R.-R. Li, G.-X. Tian, X.-C. Wang, *Angew. Chem. Int. Ed.* **2023**, *62*, e202307697.
[47] An earlier version of this manuscript was deposited to ChemRxiv on 09 December 2024: P. Roy, P. Mahto, S. Mukherjee, *ChemRxiv* **2024**, https://doi.org/10.26434/chemrxiv-2024-934x8.
[48] P. Das, D. Ghosh, S. Mukherjee, *Angew. Chem. Int. Ed.* **2024**, *63*, e202413609.
[49] A. Chakrabarty, K. Jaiswal, M. De, S. Mukherjee, *Org. Chem. Front.* **2024**, *11*, 5107–5115.
[50] S. Kezuka, K. Kanemoto, R. Takeuchi, *Tetrahedron Lett.* **2004**, *45*, 6403–6406.
[51] S. Ghosh, A. Changotra, D. A. Petrone, M. Isomura, E. M. Carreira, R. B. Sunoj, *J. Am. Chem. Soc.* **2023**, *145*, 2884–2900.
[52] M. Isomura, D. A. Petrone, E. M. Carreira, *J. Am. Chem. Soc.* **2019**, *141*, 4738–4748.
[53] D. A. Petrone, M. Isomura, I. Franzoni, S. L. Rössler, E. M. Carreira, *J. Am. Chem. Soc.* **2018**, *140*, 4697–4704.
[54] T. Sawano, M. Kobayashi, M. Ishikawa, E. Ishikawa, R. Takeuchi, *Org. Chem. Front.* **2025**, *12*, 1080–1085.
[55] S. L. Rössler, D. A. Petrone, E. M. Carreira, *Acc. Chem. Res.* **2019**, *52*, 2657–2672.
[56] Deposition Number 2426343 (for **9**) contains the supplementary crystallographic data for this paper. These data are provided free of charge by the joint Cambridge Crystallographic Data Centre and Fachinformationszentrum Karlsruhe Access Structures service.
[57] S. Krautwald, E. M. Carreira, *J. Am. Chem. Soc.* **2017**, *139*, 5627–5639.

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